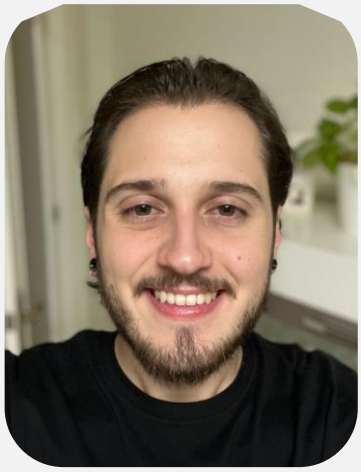


CONTACT

rafaroman182@gmail.com

+34 665 195 230

Seville, Spain

rafaroman18.github.io

github.com/rafaroman18

GENERAL SKILLS

- Data Visualization
- Deep Learning
- Machine Learning
- Agile Methodologies
- Data Processing
- ETL Pipelines
- Generative Models (GANs)
- Relational & NoSQL Databases

TECHNICAL SKILLS

Languages: Python, Pandas, PyTorch, Scikit-Learn, Keras, TensorFlow, Matplotlib, C#, Java

Databases: Spark, Hadoop, MongoDB, Powercenter, QLIK, Oracle, Teradata, Microsoft SQL Server, SQL, NO-SQL

Cloud: AWS, GCP, Azure
Other: Git, Docker, ElasticSearch

LANGUAGES

Spanish – **Native**
English – **B2 (Cambridge)**

CERTIFICATIONS

B2 First (FCE) - Cambridge
UiPath RPA Associate
AI & ML Bootcamp (Udemy, 2026)
ChatGPT & AI (Fundae, 2026)

Rafael Román Aguilar

Data Scientist · Generative AI & Deep Learning

Data Scientist with experience building **Deep Learning**, **Machine Learning** and **generative AI** solutions in Python. Skilled at analyzing and optimizing data flows, and currently designing, validating and deploying AI models in backend environments at FIDESOL. Adaptable, professional and persistent.

EDUCATION

Master's in Data Engineering and Science

National Distance Education University (UNED)
10/2022 – 06/2026

Bachelor's in Computer Engineering – Computing

School of Engineering (Universidad de Cádiz) – **Average Grade: 7.1/10**
09/2016 – 03/2021

EXPERIENCE

Data Scientist – FIDESOL

12/2024 – Present — Granada, Spain

AI research and model design, including evaluation and validation. Defined AI solution **architectures** and deployed models in **backend environments** using Python and Django REST Framework. Delivered **generative AI** virtual patients for medical training (UNIR), **LSTM** demand forecasting, heat stress detection with wearables and a smart farming platform with **YOLO** pest detection.
Technologies: Python, Django, Docker, TensorFlow, PyTorch, LLMs, Azure DevOps

PLSQL Developer – Minsait

08/2022 – 12/2024 — Sevilla, Spain

Maintained and developed the processes that feed the **databases** of a banking client. Detected and fixed incidents, analyzed **data flows** and built transformations over **large data volumes**.
Technologies: Powercenter, QLIK, OBIEE, SQL Server, Oracle, Teradata

Data Scientist – NASSAT

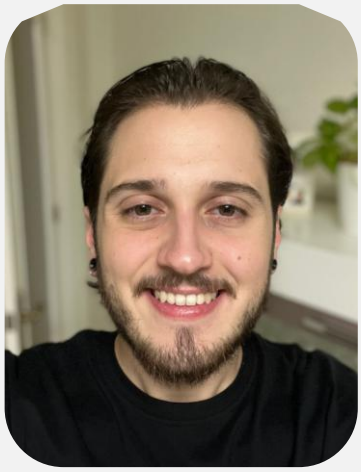
03/2022 – 04/2022 — Mijas, Spain

Deep Learning model built with **PyTorch** to predict wildfires and assign a risk index to each zone, viewable in a virtual reality headset, working with geographic and historical data (.shp, .geoJSON).
Technologies: Python, PyTorch, Pandas, NumPy

DevOps Developer – Sopra Steria

04/2021 – 03/2022 — Sevilla, Spain

Database management, website deployment with **Docker** and script monitoring with **OpenShift** for a major Andalusian client.
Technologies: Java, Python, Docker, ElasticSearch



Rafael Román Aguilar

Data Scientist · Generative AI & Deep Learning

Data Scientist with experience building **Deep Learning**, **Machine Learning** and **generative AI** solutions in Python. Skilled at analyzing and optimizing data flows, and currently designing, validating and deploying AI models in backend environments at FIDESOL. Adaptable, professional and persistent.

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ChatGPT & AI (Fundae, 2026)

PROJECTS

Master's Thesis: GANs for Breast Cancer Detection

github.com/rafaroman18/BreastCancerDL

Studied whether a **StyleGAN** trained on only 260 thermal images can generate samples that improve a breast cancer classifier. Bias-aware design with **400 StyleGAN models**, 5 augmentation settings and **5-fold cross-validation** repeated 10 times. Adding about 50% synthetic images gave the best trade-off in recall.

SmartCrops: Smart Farming Portal (FIDESOL)

Portal where users consult information about their plot: alerts, olive production forecasts and fertility maps built from **satellite**, weather, soil and water data, plus a **digital twin**. A pest module uses a **YOLO** computer vision model to detect insects and parasites and alert when a critical threshold is exceeded. Worked on **infrared satellite data** research with soil and erosion indices, and on YOLO training and testing.

Heat Stress Detection with a Smart Wristband (FIDESOL)

Prototype to detect heatstroke in workers in extreme environments. Trained a **Machine Learning** model (**decision trees**) on wristband data. When heat stress is detected, an alert is sent to a control center that logs every incident, and the wristband signals with sound and light that the worker must rest and notify a supervisor. The prototype also included a **text-to-SQL** model so company staff can query information in natural language without knowing SQL.

Bus Passenger Demand Forecasting (FIDESOL)

Predictive model of daily passenger volume so a bus company can plan extra buses in advance. Built an **LSTM** network and a **regression** model, the LSTM delivering the best results. Bus data and predictions are stored in a database and the client can **retrain** the models. Responsible for data analysis and cleaning and model design.

PUBLICATIONS

RITA: LLM Evaluation with Hallucination Taxonomies

JNIC 2025 proceedings (PDF, p. 613)

L. Vigoya, C. Fernández, C. Alba, **Rafael Román**, R. Morillas. X National Cybersecurity Research Conference (JNIC), University of Zaragoza, 2025. Methodology to evaluate LLM responses with a hallucination taxonomy, validated on GPT-4o-mini, Gemini and DeepSeek.